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GitHub Link	https://github.com/bhanuprakash-420/MLA0403-DL

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING
SLOT : A

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I P.Bhanu Prakash(192425263) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Name of the student: P.Bhanu Prakash

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 02

Total Marks: 20

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

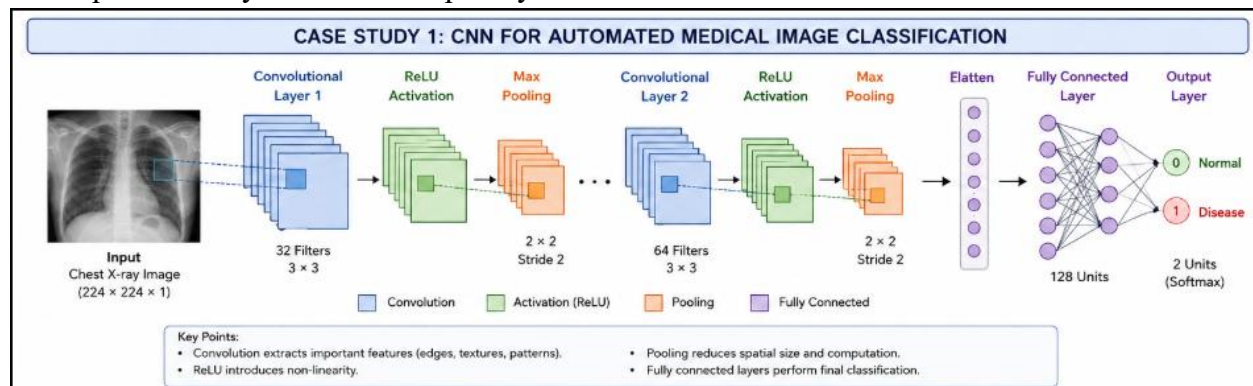
Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Answer:

1. Proposed CNN Architecture

The chest X-ray problem is well suited to a convolutional neural network because the important evidence is spatial. Instead of treating every pixel as an unrelated value, the CNN learns local patterns and combines them into increasingly meaningful representations. The proposed pipeline is: Input X-ray → normalization → convolution blocks → pooling → global feature aggregation → dropout → fully connected/output layer.



2. Why CNNs, Convolution and Filters?

CNNs are motivated by the fact that nearby pixels have meaningful relationships. A filter slides across the image and computes a weighted response for each local region, producing a feature map. Early filters commonly learn simple intensity transitions such as edges. Deeper filters combine earlier responses into textures, shapes and larger anatomical patterns. This learned hierarchy is more flexible than manually specifying rules for disease appearance.

3. Pooling and Classification

Pooling reduces the spatial dimensions of feature maps while retaining strong responses. Max pooling is useful for reducing computation and providing limited tolerance to small shifts.

However, aggressive pooling can remove subtle medical information, so downsampling should be controlled. Global average pooling is a useful final feature step because it summarizes each learned feature map without introducing the very large parameter count of a dense flattened layer.

The final fully connected layer combines these learned summaries. For binary classification, a sigmoid output can represent the estimated disease probability, or a two-node softmax can provide normal and disease scores. The model output should be interpreted as a classification result requiring appropriate clinical validation, not as an independent diagnosis.

4. Parameter Sharing and Efficiency

Parameter sharing is a major reason CNNs are practical for images. For a 224×224 grayscale input connected directly to 128 dense neurons, the first layer would need $224 \times 224 \times 128 = 6,422,528$ weights. In contrast, a convolution with 32 filters of size 3×3 needs only $3 \times 3 \times 32 = 288$ weights, plus biases. Those 32 filters are reused at every spatial position.

This reuse lowers memory and training cost and also improves translation-related feature detection. A filter that learns a useful edge or texture near the center of one X-ray can detect the same type of pattern near another location. The model therefore learns reusable visual detectors instead of separate parameters for every pixel position.

5. Regularization to Reduce Overfitting

- Data augmentation: use realistic small rotations, translations, scaling and mild brightness changes. Transformations must preserve clinically meaningful structure.
- Dropout: deactivate selected classifier activations during training so the network does not depend too strongly on a small group of features.
- Batch normalization: stabilize the distribution of intermediate activations and support more reliable optimization.
- Early stopping and validation monitoring: stop when validation performance stops improving rather than allowing the network to memorize training images.
- Balanced evaluation: report precision, recall, F1-score and the confusion matrix in addition to accuracy, especially when the two classes are not equally represented.

6. AlexNet or ResNet?

AlexNet is an important historical CNN, but its architecture is comparatively shallow and uses a large fully connected component. ResNet is deeper and uses shortcut connections that help information and gradients move through the network. This makes deeper feature learning easier to optimize and reduces the practical impact of vanishing gradients.

ResNet is therefore the better choice for this medical application. Chest X-rays contain subtle patterns that can benefit from progressively richer representations, while a compact ResNet variant can avoid unnecessary computational cost. Residual connections also provide a strong foundation for transfer learning when a suitable pretrained model is available.

7. Final Design Decision

The recommended system combines standardized X-ray input, convolutional residual feature learning, controlled pooling, global average pooling, dropout and a binary output. Its main strengths are automatic feature extraction, parameter efficiency and better control of overfitting. Independent testing is essential because a model can learn scanner-specific or hospital-specific patterns. The final system should support qualified clinical decision-making rather than replace it.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

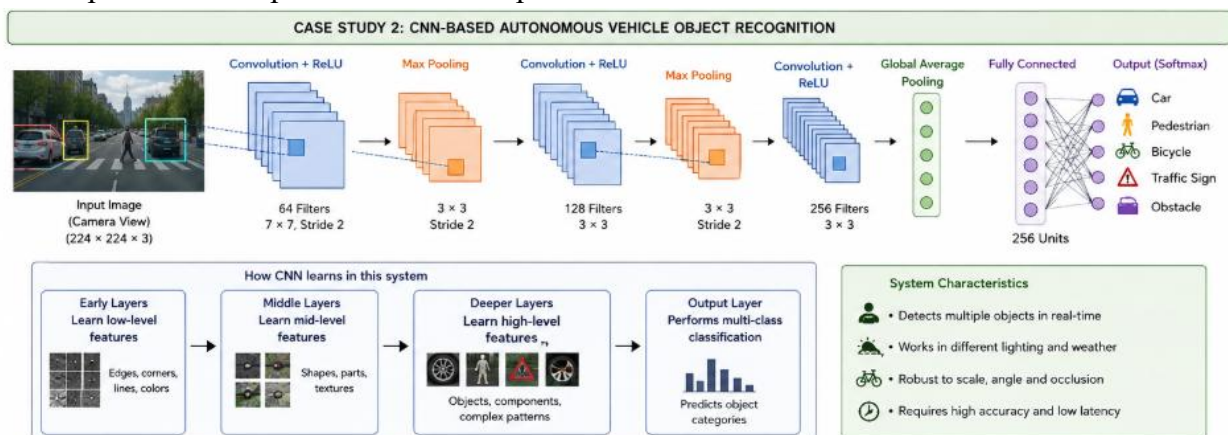
Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional filters progressively learn **low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

Answer:

1. CNN Architecture for the Vehicle Vision System

An autonomous vehicle camera produces images under changing illumination, viewpoint, distance and road conditions. The CNN must therefore learn features that remain useful when the same object appears at different sizes or positions. A suitable pipeline is: Camera frame → preprocessing → convolution → activation/normalization → downsampling → residual blocks → compact feature representation → output.



2. Purpose of the Main Layers

Convolutional filters inspect local neighborhoods and generate feature maps. Early layers learn edges, corners and brightness transitions. Middle layers combine these responses into parts such

as wheel-like shapes, sign borders or human contours. Deeper layers integrate the parts into high-level representations of vehicles, pedestrians, bicycles and other obstacles.

3. Parameter Sharing for Real-Time Vision

The same road feature can appear anywhere in a camera frame. CNN parameter sharing applies one learned filter at many locations instead of learning a separate detector for every position.

This greatly reduces the number of trainable parameters compared with a fully connected image model. It also helps the network recognize an object feature when its location changes from one frame to another.

Lower parameter counts reduce memory requirements and make computation more manageable. More importantly, shared filters learn reusable patterns, which is valuable in a moving vehicle where cars, people and signs continuously enter and leave different parts of the camera view.

4. Overfitting and Recommended Regularization

Overfitting is especially dangerous in autonomous vision because a model can appear accurate on familiar roads while failing under new conditions. For example, it may associate a class with a particular background, camera exposure or weather pattern instead of learning the object itself. The training set should therefore contain diverse roads, viewpoints, distances, lighting and environmental conditions.

- Data augmentation: vary brightness, contrast, scale, crop and translation using realistic transformations. Night and weather variation should be simulated without destroying object identity.
- Early stopping: monitor a separate validation environment and stop before continued training becomes memorization.

5. AlexNet versus ResNet

AlexNet is historically important but relatively shallow and relies more heavily on fully connected layers. ResNet uses shortcut connections, making deeper networks easier to optimize. ResNet is better suited to changes in object scale, viewpoint and appearance. A compact model can keep the classifier efficient.

7. Final Recommendation

ResNet is the recommended architecture. Its shortcut connections support deeper feature learning, while convolutional parameter sharing keeps feature extraction efficient. The final pipeline is camera capture → preprocessing → residual CNN → object representation → classification/detection output → confidence-aware vehicle logic. AlexNet is a useful historical baseline, but ResNet better combines feature learning, gradient flow and flexible model sizing.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
